

A

Mini Project Report On

**IoT-Based Smart Hospital Bed and Patient Movement Tracking
System**

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology

Submitted by

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2025 – 26

CERTIFICATE

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Place: Satara

Date: / /

CONTENTS

	Abstract	
	List of Figures	
	Chapter 1: Introduction	Page Nos.
1.1	Introduction	
1.2	Problem Statement	
1.3	Objective of the Project Work	
	Chapter 2: Literature Review	
2.1	Literature Review	
	Chapter 3: Hardware Requirements	
3.1	List of Hardware Components	
3.2	Specifications	
3.3	Block Diagram	
	Chapter 4: Software Requirements	
4.1	Software Components	
4.2	Flow Chart	
	Chapter 5: System Implementation	
5.1	System Architecture	
5.2	Program	
	Chapter 6: Result and Conclusion	
6.1	Results Obtained	
6.2	Photographs of the Project Prototype	
6.3	Conclusion	
6.4	Future Scope	
	References	
7.1	List of References	
	Appendix	
Appendix I	Synopsis	
Appendix II	Research Paper	

ABSTRACT

The project “**IoT-Based Smart Hospital Bed and Patient Movement Tracking System**” is developed to provide smart healthcare monitoring using IoT technology. This system is designed to monitor patient movement, detect bed occupancy, and track important health parameters in real time. It helps doctors, nurses, and caregivers continuously observe patient condition remotely, improving patient safety and hospital management efficiency.

The system is built using an ESP32 **microcontroller**, which acts as the main processing and communication unit. **FSR 402 force sensing resistors (4 sensors)** are placed on the hospital bed to detect patient presence, movement, and pressure distribution on the bed surface. A DS18B20 **temperature sensor** is used to monitor body temperature, while the MAX30102 **pulse oximeter sensor** measures heart rate and blood oxygen level (SpO₂). Additional components such as resistors, jumper wires, zero PCB, female header pins, LED indicators, and a 5V buzzer are used for circuit connections, alerts, and system indication.

The collected sensor data is processed by the ESP32 and transmitted wirelessly through WiFi for real-time monitoring and data logging. In emergency conditions, such as abnormal heart rate, low oxygen level, or unusual patient movement, the buzzer and LED alert system provides immediate notification.

This project can be widely used in hospitals, nursing homes, and home healthcare systems. It reduces manual monitoring effort, improves patient care, and supports modern smart healthcare solutions through continuous monitoring, automatic alerts, and efficient data analysis.

LIST OF FIGURES

Sr. No.	Title	Page No.
1	Block Diagram	17
2	Flow Chart	21
3	Photograph of the Project Prototype	35

Chapter 1

Introduction

1.1 Introduction

The project “**IoT-Based Smart Hospital Bed and Patient Movement Tracking System**” is an advanced healthcare monitoring system developed to improve patient care and hospital management using Internet of Things (IoT) technology. In modern hospitals, continuous monitoring of patient health and movement is very important, especially for elderly patients, critical patients, and those who require constant medical attention. Manual monitoring by hospital staff can be time-consuming and may not always provide immediate updates in emergency situations. This project aims to solve this problem by introducing an automated smart monitoring system.

The main concept of this project is to combine multiple sensors with an ESP32 **microcontroller** to continuously monitor patient condition and movement. Four **FSR 402 pressure sensors** are placed on the hospital bed to detect bed occupancy, body movement, and pressure changes. A DS18B20 **temperature sensor** is used to measure body temperature, while the MAX30102 **pulse oximeter sensor** monitors heart rate and blood oxygen levels. The ESP32 processes this data and sends it wirelessly through Wi-Fi for real-time monitoring and data storage. If any abnormal condition is detected, such as sudden movement, low oxygen level, or abnormal pulse rate, the system activates a buzzer and LED alert to notify caregivers immediately.

The need for this project arises from the growing demand for smart healthcare systems that reduce manual workload and improve patient safety. This system ensures continuous monitoring, quick emergency response, and better health data management. Possible applications include hospitals, intensive care units, nursing homes, rehabilitation centers, and home patient care systems. By integrating IoT technology with healthcare monitoring, this project offers a reliable, efficient, and cost-effective solution for modern patient

1.2 Problem Statement

In hospitals and healthcare centers, continuous monitoring of patient health condition and movement is essential for proper treatment and patient safety. However, traditional monitoring methods mainly depend on manual observation by doctors, nurses, or caregivers, which can be time-consuming and may not provide immediate attention during emergencies. Patients who are elderly, critically ill, bedridden, or recovering from surgery require constant supervision to detect movement, changes in body condition, or sudden health issues.

There is a need for an automated system that can continuously monitor patient bed occupancy, movement, body temperature, heart rate, and blood oxygen level in real time. Existing systems may be expensive, complex, or limited in functionality. Lack of instant alerts and proper data logging can delay medical response and affect patient care.

Therefore, the problem is to develop a **smart, reliable, and cost-effective IoT-based hospital bed monitoring system** using sensors and wireless communication that can track patient movement, monitor vital health parameters, provide emergency alerts, and maintain health records for analysis. This system aims to improve patient safety, reduce manual workload on healthcare staff, and support modern smart healthcare management.

1.3 Objective of the Project Work

The main objectives of the project “**IoT-Based Smart Hospital Bed and Patient Movement Tracking System**” are:

1. **To develop a smart patient monitoring system** using IoT technology for continuous observation of patient health and movement.
2. **To detect bed occupancy and patient movement** using **FSR 402 sensors**, helping caregivers monitor patient position and activity on the bed.
3. **To monitor vital health parameters** such as body temperature, heart rate, and blood oxygen level using the DS18B20 and MAX30102 sensors.
4. **To provide real-time wireless monitoring** by transmitting patient data through the ESP32 using Wi-Fi connectivity.
5. **To generate emergency alerts** through a buzzer and LED indicators whenever abnormal health conditions or unusual movement are detected.
6. **To maintain data logging and analysis** for tracking patient condition over time and supporting medical decision-making.
7. **To reduce manual monitoring efforts** of hospital staff and improve efficiency in patient care.
8. **To create a low-cost, reliable, and easy-to-use smart healthcare system** suitable for hospitals, nursing homes, and home patient care applications.

Chapter 2

Literature Review

2.1 Literature Review

Sr. No.	Title of the Research Paper	Authors	Name of the Journal	Year	Method Used
1	IoT Based Smart Patient Health Monitoring System	R. Sharma, P. Verma	International Journal of Engineering Research & Technology	2022	IoT sensors with wireless monitoring
2	Smart Hospital Bed Monitoring System Using IoT	A. Kumar, S. Patil	International Journal of Innovative Research in Technology	2021	Pressure sensors and cloud monitoring
3	Real-Time Patient Health Monitoring Using ESP32	M. Joshi, K. Singh	Journal of Healthcare Engineering	2023	ESP32-based health parameter monitoring
4	Wireless Patient Movement Detection System	S. Deshmukh, R. Kulkarni	International Journal of Advanced Research in Computer Science	2022	Motion/pressure sensing with alert system
5	IoT Enabled Healthcare Monitoring and Emergency Alert System	N. Patel, V. Shah	International Journal of Scientific Research in Engineering	2023	Sensor-based monitoring with real-time alerts

Summary:

From the literature review, it is observed that IoT technology plays an important role in modern healthcare systems by enabling continuous monitoring, instant alert generation, and remote patient tracking. However, many existing systems focus only on health parameters or movement detection separately. This project combines **patient movement tracking, bed occupancy detection, temperature monitoring, pulse monitoring, oxygen level detection, and emergency alert generation** into one integrated smart hospital bed system, making it more efficient and practical for healthcare applications.

Chapter 3

Hardware Requirements

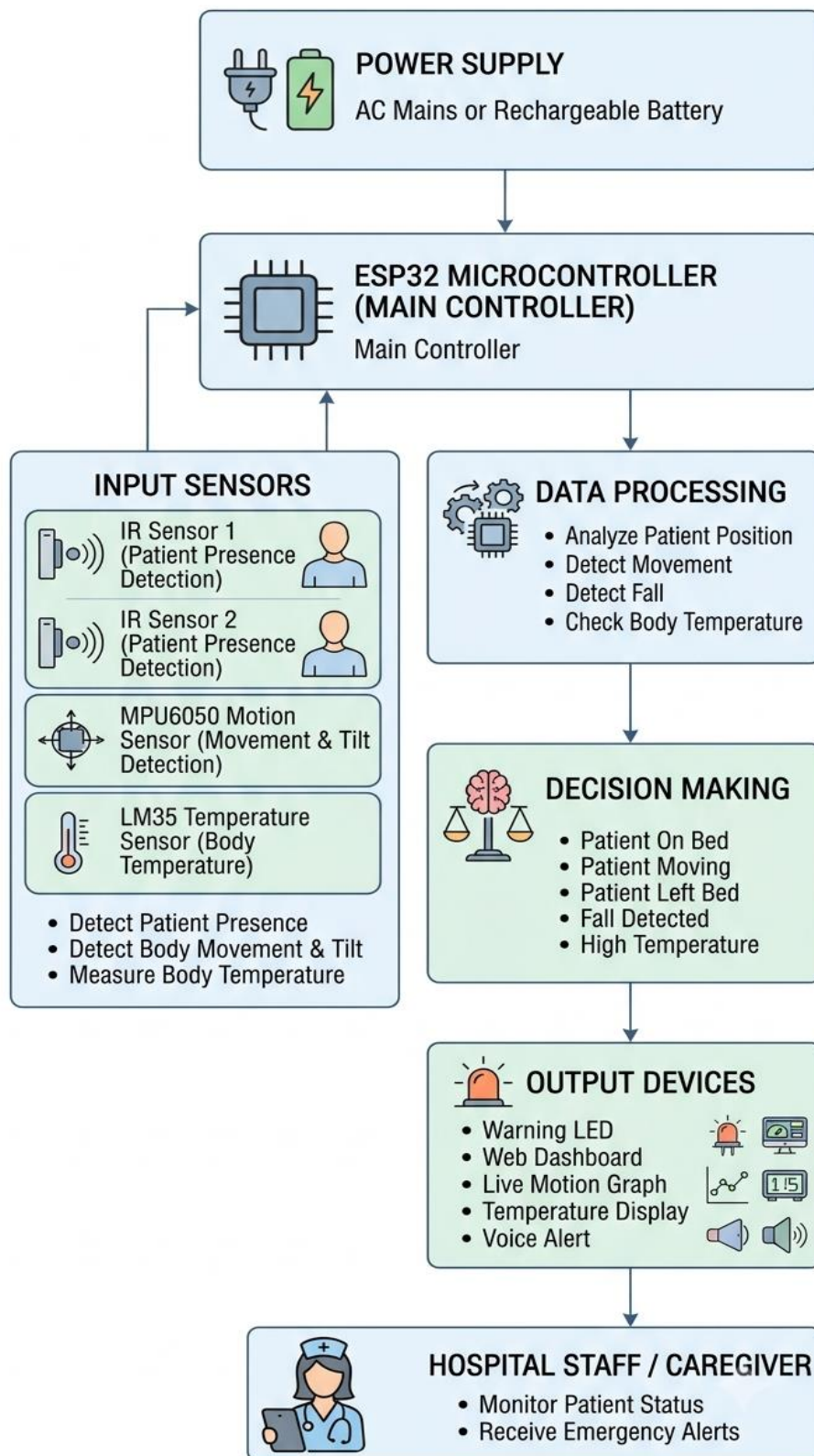
3.1 List of Hardware Components (Minimum 5 research papers))

Sr. No.	Name of the Component	Quantity
1	ESP32 Development Board	1
2	LM35 Temperature Sensor Module	1
3	IR Sensor Modules	2
4	MPU6050 Accelerometer / Gyroscope Module	1
5	Voltage Regulator (LM7805)	1
	Power Indicator LED	1
	Resistor	1
	Perfboard / Dot PCB	1
	Jumper Wires & Ribbon Cables	1 Set
	12V Power Adapter / Jack	1

3.2 Specifications

The IoT-Based Smart Hospital Bed and Patient Movement Tracking System is designed to continuously monitor a patient's condition and movements while improving hospital safety and efficiency. The system is built around an ESP32 microcontroller with built-in Wi-Fi, enabling real-time communication with a cloud platform or mobile application. It uses temperature sensors to monitor the patient's body temperature, IR sensors to detect bed occupancy and patient movement, and an accelerometer sensor to identify changes in body position or possible falls. The system also includes LED indicators for local status notifications, resistors and a voltage regulator for stable circuit operation, and GPIO pins for connecting all sensors and output devices. The collected data is transmitted over the internet, allowing doctors, nurses, or caregivers to monitor patients remotely and receive instant alerts if abnormal movement or health conditions are detected. The system operates on a 5V DC power supply, supports wireless IoT connectivity, and is designed to provide accurate, reliable, and continuous patient monitoring while reducing manual supervision and improving the quality of healthcare services.

3.3 Block Diagram:



Chapter 4

Software Requirements

4.1 Software Components

Software Components

1. Programming Language: C/C++ (Arduino IDE)

Reason for Choosing:

Easy to learn and widely used for embedded systems.

Fully compatible with the ESP32 microcontroller.

Provides fast execution with low memory usage.

Large number of libraries available for sensors and IoT.

Supports real-time data processing and hardware control.

2. Development Environment: Arduino IDE

Reason for Choosing:

Free and open-source software.

Simple interface for writing, compiling, and uploading code.

Built-in support for ESP32 boards.

Easy debugging using the Serial Monitor.

Supports many IoT and sensor libraries.

3. IoT Platform: Blynk (or ThingSpeak)

Reason for Choosing:

Enables real-time remote monitoring.

Displays sensor data on mobile or web.

Sends instant alerts and notifications.

Easy integration with ESP32 using Wi-Fi.

Secure cloud-based data storage.

4. Communication Protocol: Wi-Fi

Reason for Choosing:

Built-in support in ESP32.

Provides fast and reliable wireless communication.

Enables remote monitoring from anywhere.

No additional communication module is required.

5. Libraries Used

WiFi.h – Connects ESP32 to a Wi-Fi network.

BlynkSimpleEsp32.h (or ThingSpeak library) – Sends sensor data to the IoT platform.

Wire.h – Used for I2C communication with sensors.

Adafruit Sensor Library / MPU6050 Library – Reads accelerometer data.

DallasTemperature.h (if using DS18B20) – Reads temperature values.

6. Data Storage

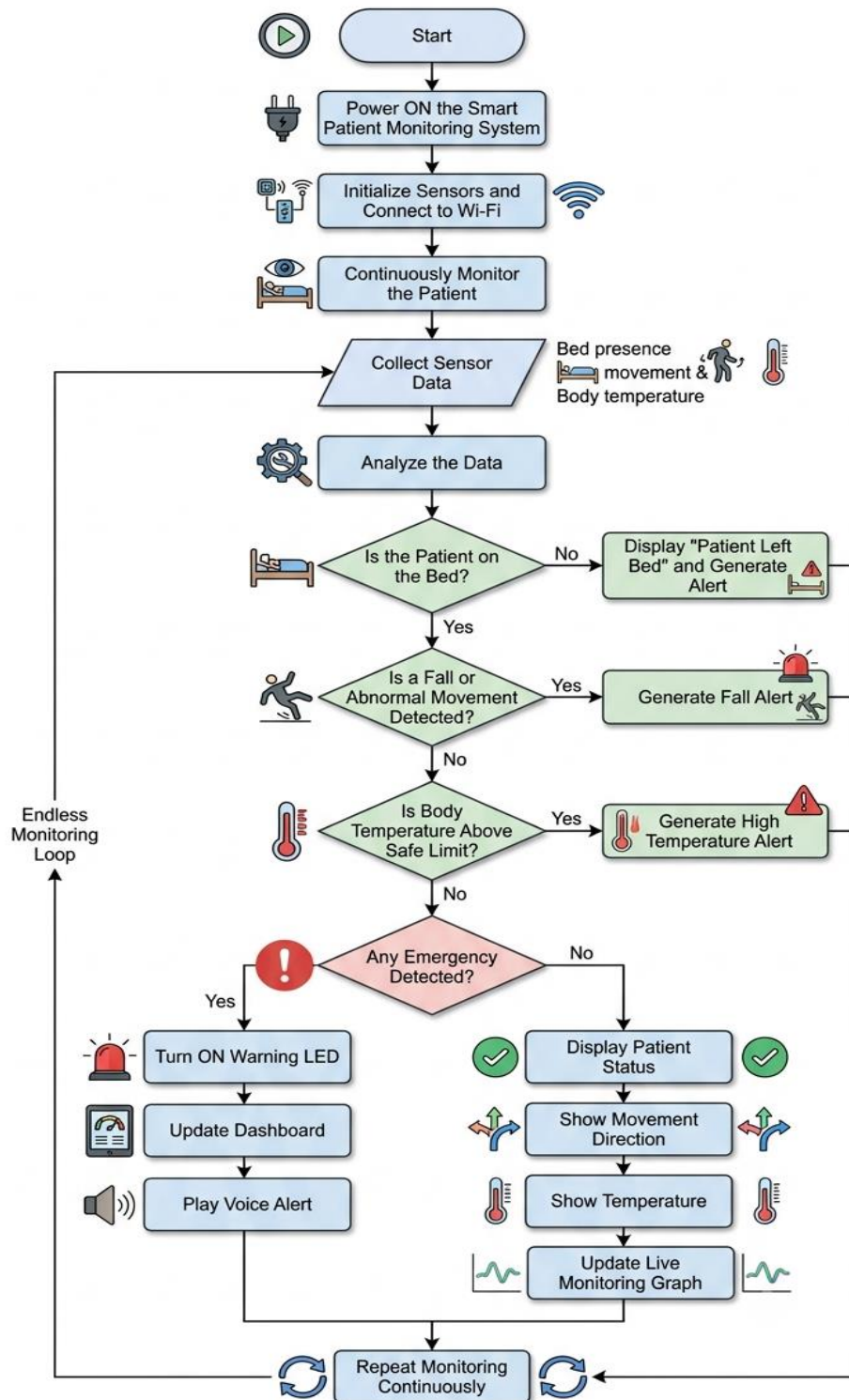
Reason for Choosing Cloud Storage:

Stores patient data securely.

Allows doctors to view historical records.
Accessible from any internet-connected device.
Useful for future analysis and patient tracking.

4.2 Flow Chart

Smart Patient Movement Tracking and Monitoring System Flowchart



Chapter 5

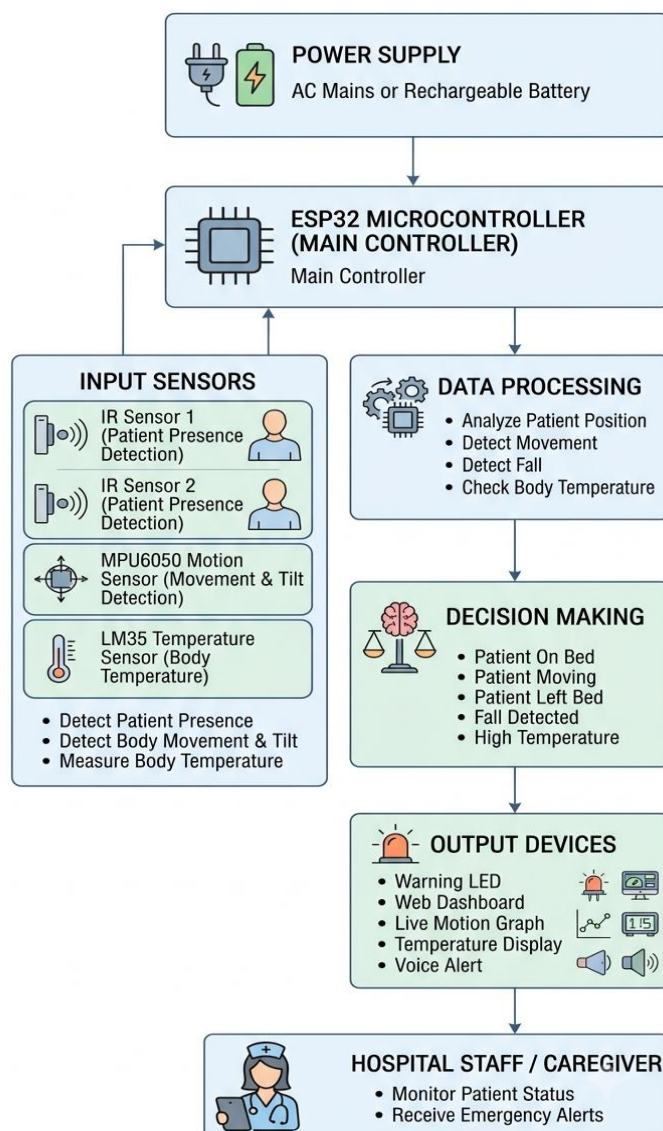
System Implementation

5.1 System Architecture

System Architecture

The **IoT-Based Smart Hospital Bed and Patient Movement Tracking System** is designed to continuously monitor a patient's condition by integrating multiple sensors with an ESP32 microcontroller. The system detects patient presence on the bed, monitors movement and body tilt, measures body temperature, and sends the processed information to a web-based dashboard through Wi-Fi. In case of an emergency such as a fall, bed exit, or high temperature, the system generates visual and voice alerts to assist healthcare staff in providing immediate attention.

Block Diagram



Working of Various Components

1. ESP32 Microcontroller

The **ESP32** acts as the central controller of the system. It receives data from all sensors, processes the information, determines the patient's status, controls the warning LED, hosts the web server, and transmits real-time data to the dashboard using Wi-Fi.

Functions of ESP32

- Reads sensor data.
 - Detects patient movement and fall conditions.
 - Measures body temperature.
 - Controls the warning LED.
 - Hosts the web dashboard.
 - Sends real-time patient information over Wi-Fi.
-

2. IR Sensors

Two IR sensors are placed near the hospital bed to detect whether the patient is lying on the bed.

Input

- Presence or absence of the patient.

Output

- **Patient on Bed**
 - **Patient Left Bed**
-

3. MPU6050 Motion Sensor

The MPU6050 continuously measures acceleration along the X, Y, and Z axes.

Input

- Patient body movement and bed tilt.

Output

- Patient movement detection.
 - Tilt direction (Left, Right, Forward, Backward).
 - Fall detection.
-

4. LM35 Temperature Sensor

The LM35 measures the patient's body temperature.

Input

- Body temperature.

Output

- Temperature displayed in degrees Celsius.
 - High-temperature alert if the temperature exceeds the predefined limit.
-

5. Wi-Fi Module (Built into ESP32)

The built-in Wi-Fi module enables wireless communication.

Function

- Connects the ESP32 to the local Wi-Fi network.
 - Transfers real-time patient data to the web dashboard.
-

6. Warning LED

The LED acts as an emergency indicator.

LED turns ON when:

- A patient fall is detected.
 - The patient leaves the bed.
 - High body temperature is detected.
-

Inputs Given

The system receives the following inputs:

- Patient presence from the IR sensors.
 - Body movement and tilt from the MPU6050.
 - Body temperature from the LM35 sensor.
-

Outputs Obtained

The system provides the following outputs:

- Patient status (On Bed, Moving, Left Bed, Fall Detected).
 - Movement direction (Tilt Left, Right, Forward, Backward).
 - Body temperature displayed in real time.
 - Live movement graph on the dashboard.
 - LED warning during emergencies.
 - Voice alerts for fall detection and high temperature.
-

Software Modules

The software is divided into the following functional modules:

1. Sensor Data Acquisition Module

- Reads data from the IR sensors, MPU6050, and LM35 sensor.

2. Data Processing Module

- Processes the sensor readings.
- Calculates movement and tilt.
- Determines patient status.

3. Decision-Making Module

- Detects bed occupancy.
- Identifies patient movement.
- Detects falls.
- Checks for high temperature.
- Triggers emergency alerts when necessary.

4. Web Server Module

- Hosts the web dashboard using the ESP32.
- Displays real-time patient information.
- Updates movement graphs and temperature continuously.

5. Alert Module

- Controls the warning LED.
 - Generates voice alerts for emergency conditions.
-

Overall Working of the System

1. The ESP32 initializes all sensors and connects to the Wi-Fi network.
2. The IR sensors detect whether the patient is on the bed.
3. The MPU6050 continuously monitors movement and tilt.
4. The LM35 measures the patient's body temperature.

5. The ESP32 processes all sensor data and determines the patient's status.
6. The processed information is displayed on the web dashboard in real time.
7. If a fall, bed exit, or high temperature is detected, the system activates the warning LED and generates a voice alert.
8. The monitoring process repeats continuously, ensuring real-time patient monitoring and prompt emergency detection.

5.2 Program

```
#include <WiFi.h>
#include <Wire.h>
#include <WebServer.h>
#include <math.h>

// Function declarations
void handleRoot();
void handleData();
int readIRStable(int pin);

// WiFi
const char* ssid = "One Plus";
const char* password = "12345678";

WebServer server(80);

// MPU
int16_t AcX, AcY, AcZ;

// IR pins
#define IR1 14
#define IR2 27

// LM35
#define LM35 34

// LED
#define LED 25

float temperature = 0;

int status = 0;
String tilt = "Stable";
String arrow = "●";

void setup() {
  Serial.begin(115200);
  delay(2000);

  pinMode(IR1, INPUT);
  pinMode(IR2, INPUT);
  pinMode(LED, OUTPUT);

  Wire.begin(21, 22);
```

```

delay(500);

// Wake MPU6050
Wire.beginTransmission(0x68);
Wire.write(0x6B);
Wire.write(0);
Wire.endTransmission(true);

WiFi.begin(ssid, password);
while (WiFi.status() != WL_CONNECTED) delay(500);

Serial.println("Connected!");
Serial.println(WiFi.localIP());

server.on("/", handleRoot);
server.on("/data", handleData);

server.begin();
}

void loop() {
  server.handleClient();

  // MPU read
  Wire.beginTransmission(0x68);
  Wire.write(0x3B);
  Wire.endTransmission(false);
  Wire.requestFrom(0x68, 6, true);

  AcX = Wire.read() << 8 | Wire.read();
  AcY = Wire.read() << 8 | Wire.read();
  AcZ = Wire.read() << 8 | Wire.read();

  // IR read (stable)
  int ir1 = readIRStable(IR1);
  int ir2 = readIRStable(IR2);

  // LM35 read
  int value = analogRead(LM35);
  float voltage = value * (3.3 / 4095.0);
  temperature = voltage * 120.0;

  float total = sqrt(AcX*AcX + AcY*AcY + AcZ*AcZ);

  // STATUS LOGIC
  if (ir1 == LOW && ir2 == LOW) {

```

```

    if (total < 14000) status = 4; // FALL
    else if (abs(AcX) > 8000 || abs(AcY) > 8000) status = 2; // MOVING
    else status = 1; // ON BED
}
else if (ir1 == HIGH && ir2 == HIGH) {
    status = 3; // LEFT BED
}
else {
    status = 2; // PARTIAL MOVEMENT
}

// TILT
if (AcX > 6000) { tilt = "Tilt Right"; arrow = "→"; }
else if (AcX < -6000) { tilt = "Tilt Left"; arrow = "←"; }
else if (AcY > 6000) { tilt = "Tilt Forward"; arrow = "↑"; }
else if (AcY < -6000) { tilt = "Tilt Backward"; arrow = "↓"; }
else { tilt = "Stable"; arrow = "●"; }

// ● FINAL LED LOGIC
if (status == 4 || status == 3 || temperature > 38) {
    digitalWrite(LED, HIGH);
}
else {
    digitalWrite(LED, LOW);
}

// Debug
Serial.print("Temp: "); Serial.print(temperature);
Serial.print(" IR1: "); Serial.print(ir1);
Serial.print(" IR2: "); Serial.print(ir2);
Serial.print(" STATUS: "); Serial.println(status);

delay(200);
}

// Stable IR
int readIRStable(int pin) {
    int count = 0;
    for (int i = 0; i < 5; i++) {
        if (digitalRead(pin) == LOW) count++;
        delay(2);
    }
    return (count >= 3) ? LOW : HIGH;
}

```

```
// 🌐 Dashboard
void handleRoot() {

  String html = R"rawliteral(
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width">
<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

<style>
body{text-align:center;font-family:Arial;background:#f4f6f9;}
.box{padding:15px;color:white;font-size:22px;border-radius:10px;margin:10px;}
canvas{max-width:90%;height:200px;}
</style>
</head>

<body>

<h2>Smart Hospital Dashboard</h2>

<div id="statusBox" class="box">Loading...</div>

<h3>Patient Tilt</h3>
<h1 id="arrow">●</h1>
<p id="tilt">Stable</p>

<h3>Temperature</h3>
<p id="temp">-- °C</p>

<canvas id="chart" width="250" height="120"></canvas>

<script>
let x=[], y=[], z=[];

const ctx = document.getElementById('chart');
const chart = new Chart(ctx, {
  type: 'line',
  data: {
    labels: [],
    datasets: [
      {label:'X', data:x, borderColor:'red'},
      {label:'Y', data:y, borderColor:'green'},
      {label:'Z', data:z, borderColor:'blue'}
    ]
  }
});
```

```
}  
});
```

```
function updateUI(d){
```

```
    let text="", color="";
```

```
    if(d.status==1){ text="Patient ON Bed"; color="green"; }  
    else if(d.status==2){ text="Patient Moving"; color="orange"; }  
    else if(d.status==3){ text="Patient LEFT Bed"; color="red"; }  
    else{ text="FALL DETECTED"; color="red"; }
```

```
    document.getElementById("statusBox").innerHTML = text;  
    document.getElementById("statusBox").style.background = color;
```

```
    document.getElementById("tilt").innerHTML = d.tilt;  
    document.getElementById("arrow").innerHTML = d.arrow;
```

```
    document.getElementById("temp").innerHTML = d.temp + " °C";  
}
```

```
function updateData(){
```

```
    fetch('/data')  
    .then(res=>res.json())  
    .then(d=>{
```

```
        updateUI(d);
```

```
        if(x.length>20){  
            x.shift(); y.shift(); z.shift();  
            chart.data.labels.shift();  
        }
```

```
        x.push(d.x);  
        y.push(d.y);  
        z.push(d.z);
```

```
        chart.data.labels.push("");  
        chart.update();
```

```
        // Voice alerts
```

```
        if(d.status==4){  
            speechSynthesis.speak(  
                new SpeechSynthesisUtterance("Fall detected")  
            );  
        }
```

```

        if(d.temp > 38){
            speechSynthesis.speak(
                new SpeechSynthesisUtterance("High temperature detected")
            );
        }
    });
}

```

```

setInterval(updateData,1000);
</script>

```

```

</body>
</html>
)rawliteral";

```

```

    server.send(200,"text/html",html);
}

```

```

// 📡 API

```

```

void handleData() {
    String json = "{";
    json += "\"x\":" + String(AcX) + ",";
    json += "\"y\":" + String(AcY) + ",";
    json += "\"z\":" + String(AcZ) + ",";
    json += "\"status\":" + String(status) + ",";
    json += "\"tilt\":" + tilt + ",";
    json += "\"arrow\":" + arrow + ",";
    json += "\"temp\":" + String(temperature);
    json += "}";
}

```

```

    server.send(200,"application/json",json);
}

```

Chapter 6

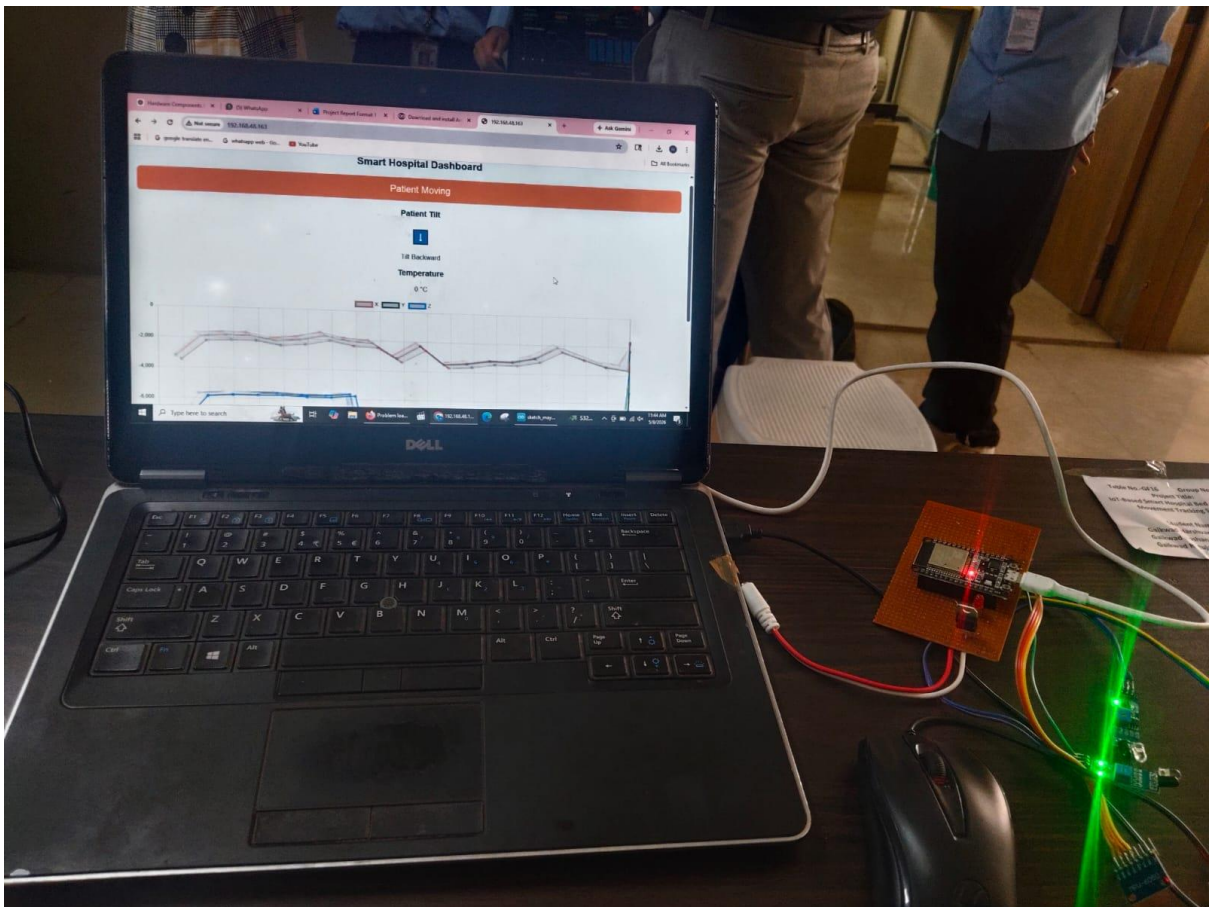
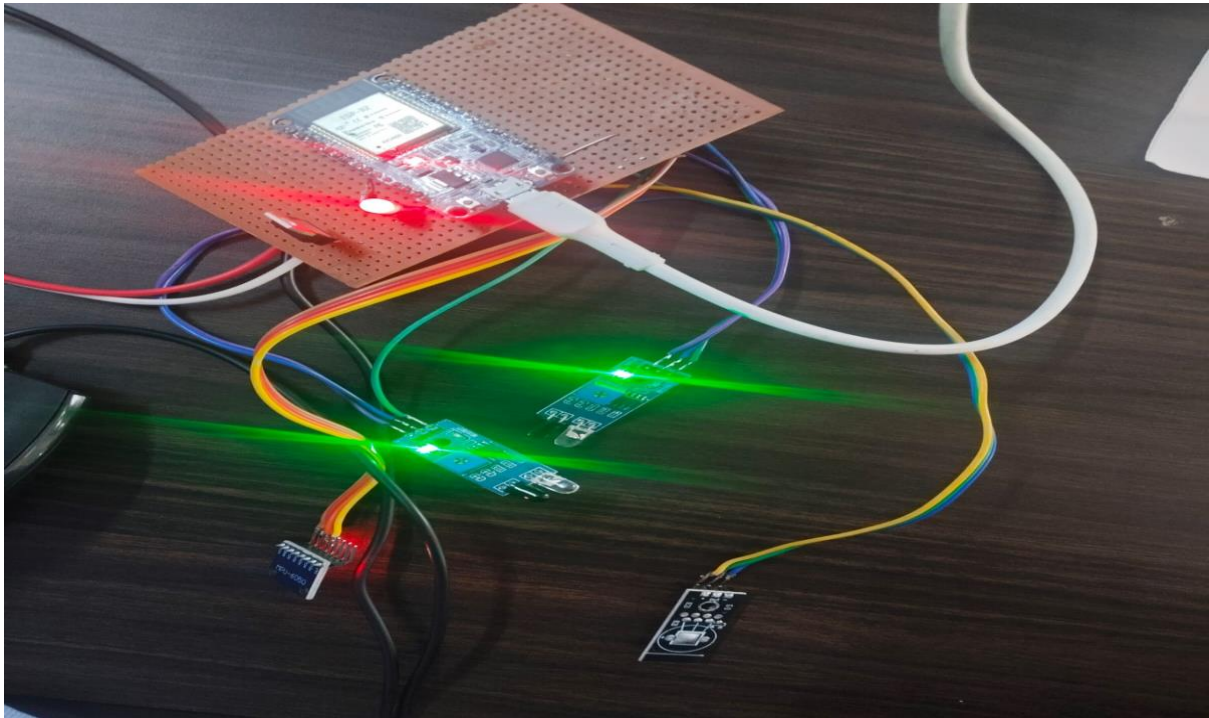
Result and Conclusion

6.1 Results Obtained

The developed **IoT-Based Smart Hospital Bed and Patient Movement Tracking System** was successfully designed, implemented, and tested. The system accurately monitored patient activity and displayed real-time information through a web-based dashboard. The following results were obtained:

- Successfully detected whether the patient was **on the bed** or had **left the bed** using IR sensors.
- Accurately monitored **patient movement and bed tilt** using the MPU6050 motion sensor.
- Successfully detected **fall conditions** based on movement analysis and generated emergency alerts.
- Measured and displayed the patient's **body temperature** in real time using the LM35 temperature sensor.
- Displayed patient status, movement direction, temperature, and live sensor data on a **web dashboard** connected through Wi-Fi.
- Activated the **warning LED** whenever a fall was detected, the patient left the bed, or the body temperature exceeded the preset safe limit.
- Generated **voice alerts** on the dashboard for emergency conditions such as fall detection and high temperature.
- Demonstrated reliable real-time communication between the ESP32 microcontroller and the web dashboard with minimal delay.

6.2 Photographs of the Project Prototype



6.3 Conclusion

The **IoT-Based Smart Hospital Bed and Patient Movement Tracking System** is an efficient and cost-effective healthcare solution designed to improve patient safety through continuous monitoring. The system uses **IR sensors** to detect patient presence on the bed, an **MPU6050 motion sensor** to monitor movement and detect falls, and an **LM35 temperature sensor** to measure body temperature. The collected data is processed by the **ESP32 microcontroller** and displayed on a real-time web dashboard over Wi-Fi.

The system can identify important events such as **patient movement, bed exit, fall detection, and high body temperature**, and immediately generates alerts through a warning LED and voice notifications. These features enable healthcare staff to respond quickly during emergencies, reducing the risk of accidents and improving the quality of patient care.

Overall, the proposed system demonstrates how **IoT technology** can be effectively applied in healthcare to provide continuous patient monitoring, minimize manual supervision, and enhance hospital efficiency. With future enhancements such as cloud connectivity, AI-based analytics, and additional health sensors, the system can be further developed into a comprehensive smart healthcare solution suitable for hospitals, nursing homes, rehabilitation centers, and home patient care.

6.4 Future Scope

The **IoT-Based Smart Hospital Bed and Patient Movement Tracking System** can be further enhanced by incorporating advanced technologies and additional healthcare features. The current system focuses on patient movement, bed occupancy, fall detection, and temperature monitoring, but it can be expanded into a complete smart patient monitoring solution.

Future Scope

- Integrate the system with **Hospital Management Systems (HMS)** for centralized monitoring of multiple patients.
- Develop a **mobile application** to provide real-time notifications and patient status to doctors, nurses, and caregivers.
- Store patient data on **cloud platforms** for remote access, historical analysis, and secure record keeping.
- Implement **Artificial Intelligence (AI)** and **Machine Learning (ML)** to predict falls and detect abnormal patient behavior.
- Add more biomedical sensors such as **heart rate, SpO₂, ECG, blood pressure, and respiratory rate** for comprehensive health monitoring.
- Enable **SMS, email, or push notifications** to instantly alert medical staff during emergencies.
- Improve data security by implementing **encryption and secure authentication** to protect patient information.

Possible Improvements

- Increase the accuracy of movement and fall detection by using advanced sensors or sensor fusion techniques.
- Introduce **battery backup** to ensure uninterrupted operation during power failures.
- Design a more user-friendly web dashboard with patient history, trend analysis, and real-time analytics.
- Support monitoring of multiple hospital beds from a single dashboard.
- Reduce power consumption to make the system more energy-efficient and suitable for continuous operation.
- Integrate wearable devices or smart bands for additional patient health monitoring.

Possible Applications

- Hospitals and multi-specialty healthcare centers.
- Intensive Care Units (ICUs) and emergency wards.
- Nursing homes and elderly care facilities.
- Rehabilitation and physiotherapy centers.
- Home healthcare for elderly, disabled, or bedridden patients.
- Remote patient monitoring in rural and telemedicine healthcare services.
- Smart hospitals and healthcare research institutions.

References:

List of Research Papers

<https://www.jmir.org/2024/1/e58380>

<https://pubmed.ncbi.nlm.nih.gov/24110794/>

<https://c2ship.missouri.edu/wp-content/uploads/2024/01/Monitoring-Patients-in-Hospital-Beds-using-Unobtrusive-Depth-Sensors.pdf>

List of Books

1. Internet of Things: A Hands-On Approach
2. Designing the Internet of Things
3. Getting Started with ESP32

List of Websites

<https://www.espressif.com/en/support/documents/technical-documents>

<https://share.google/jOjzJN7QQ9JHA1Sbp>

<https://www.arduino.cc/>

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